Mastersing Essesential Docker Commands

1. Docker Pull

- Used to download/pull a Docker Image from a Docker Registry to our local system

```bash
docker pull nginx
```

```
Docker Hub
    ↓ pull
Your Computer
    ↓
Nginx Image
```

2. Docker Push

- Used to **upload/push** a Docker Image from your local system to **Docker Hub/Registry**

Bash

```
docker push username/image-name
```

```
Your Computer
    ↓ push
Docker Hub
```

3. Docker Run

- Used to **create and run a** container from an Image.

```Bash
docker run image-name
```

Common options

- Interactive :-
  - “-i” stands for interactive
  - Keeps interaction with the container through the terminal
- Detached :-
  - “-d” stands for detached
  - Runs the container as a **background process**
  - Your terminal remains available for other commands
- Port Mapping :-
  - “-p” stands for port mapping
  - Maps a port on the host machine to a port inside the **container**
- Name Mapping :-
  - --name :- assigns a name to the container

interactive

`-i` → **Interactive**

- `i` stands for **interactive**.
- Keeps interaction with the container through the terminal.

Detached

`-d` → **Detached mode**

- Runs the container as a **background process**.
- Your terminal remains available for other commands.

Example:

Bash

```
docker run -d nginx
```

Example:

Bash

```
docker run -d nginx
```

`-p` → **Port mapping**

- Maps a port on the **host machine** to a port inside the **container**.

```bash
docker run -p 8080:80 nginx
```

Meaning:

```
Host                Container
8080      ──────►     80
```

So you can access the container's application through port `8080` on your machine.

`--name`

- Assigns a name to the container.

Example:

Bash

```
docker run -d -p 8080:80 --name my-nginx nginx
```

Here:

- `my-nginx` = container name
- `nginx` = image name

4. Docker Stop

- Used to **stop a running container**.
- You can specify:
  - Container ID
  - Container name

Example:

‹/› **Bash**

```
docker stop my-nginx
```

5. Docker Start

- Used to **start an existing stopped container**.
- Specify the container:
  - ID
  - Name

Example:

```bash
docker start my-nginx
```

Important distinction

```
docker run
    ↓
Creates + starts a NEW container

docker start
    ↓
Starts an EXISTING stopped container
```

6. Docker rm

- Used to **remove a stopped container**.
- Example:

```bash
docker rm my-nginx
```

- It removes the container itself.
- It does **not** mean removing the Docker image.

7. Docker rmi

- Used to **remove a Docker image from local storage**.
- Images consume disk space.
- You can remove images that you no longer need.

Example:

Bash

```
docker rmi image-name
```

or:

Bash

```
docker rmi image-id
```

Remember

```
docker rm
    ↓
Remove CONTAINER

docker rmi
    ↓
Remove IMAGE
```

8. Docker ps

- Used to list **currently running containers**.

Bash

```
docker ps
```

Example output can contain:

```
CONTAINER ID
IMAGE
COMMAND
STATUS
PORTS
NAMES
```

If no containers are running, you may get an empty result.

9. Docker ps -a

- Shows **all containers**:
  - Running containers
  - Stopped containers

Bash

```
docker ps -a
```

Easy way to remember

```
docker ps
    ↓
Running containers

docker ps -a
    ↓
ALL containers
```

10. Docker Images

- Lists the Docker images stored on your local machine.

```bash
docker images
```

You can see things such as:

```
REPOSITORY
TAG
IMAGE ID
CREATED
SIZE
```

For example:

```
nginx
hello-world
my-spring-app
```

11. Docker exec -it

- Used to **access a running container from the terminal**.
- `exec` → execute a command inside the container.
- `-i` → interactive.
- `-t` → allocate a terminal.

Typical command:

Bash

```
docker exec -it container-name bash
```

Example:

Bash

```
docker exec -it my-nginx bash
```

This gives you a terminal inside the running container.

Conceptually:

```
Your Terminal
      |
      | docker exec -it
      ▼
Running Container
      |
      ▼
Container Terminal
```

12. Docker build -t

- Used to **build a Docker image from a Dockerfile**.

Example:

```bash
docker build -t username/my-app .
```

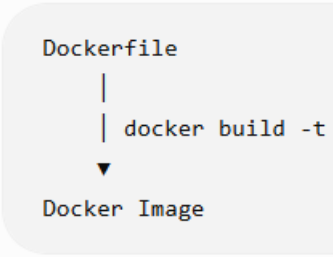
`-t`

- `t` is used to **tag/name the image**.

`.`

- The dot represents the **current directory**.
- Docker looks in the current directory for the Dockerfile and build context.

Flow:



13. Docker logs

- Used to view the **logs/output of a container**.
- Very useful when an application inside the container isn't behaving correctly.

Example:

```bash
docker logs my-nginx
```

Useful for troubleshooting:

```
Application not working
        ↓
docker logs
        ↓
Look for errors
        ↓
Understand the problem
```

14. Docker Inspect

- Provides **detailed information about a Docker object**, such as a container.
- Example:

Bash

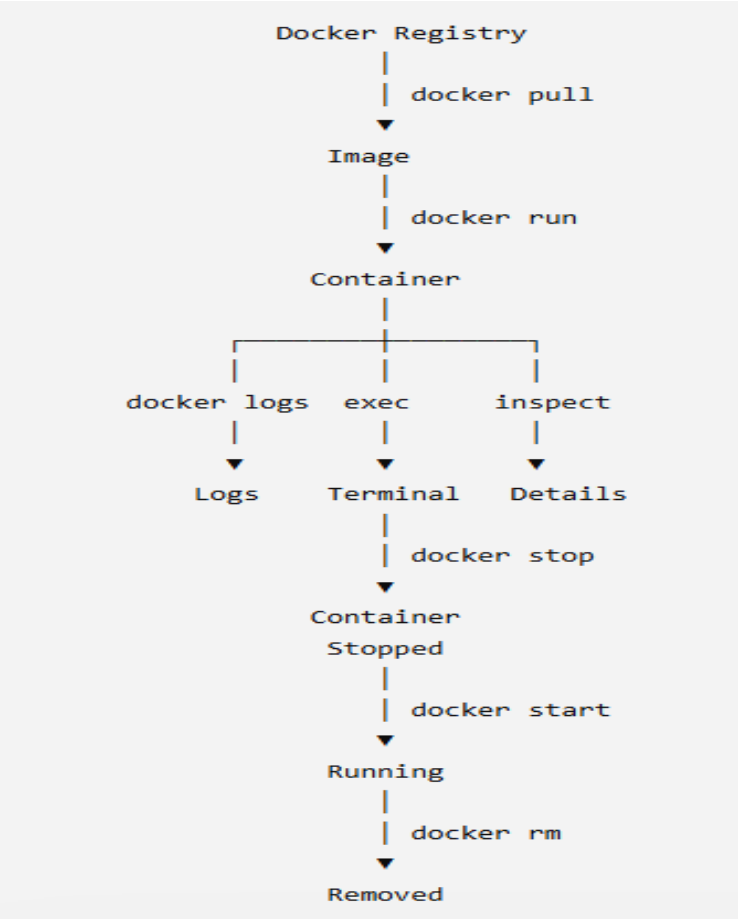
```
docker inspect my-nginx
```

It can help you investigate details about the container's configuration and environment.

Essential Commands

Command	Purpose
`docker pull`	Download image from registry
`docker push`	Upload image to registry
`docker run`	Create and run a new container
`docker stop`	Stop a running container
`docker start`	Start an existing stopped container
`docker rm`	Remove a container
`docker rmi`	Remove an image
`docker ps`	Show running containers
`docker ps -a`	Show all containers
`docker images`	List local images
`docker exec -it`	Enter/access a running container
`docker build -t`	Build an image from Dockerfile
`docker logs`	View container logs
`docker inspect`	View detailed object information

Common Docker lifecycle



#	COMMAND	DESCRIPTION	EXAMPLE	USE CASE
1	docker pull	Pull an image from a Docker registry.	`docker pull nginx`	Download an image from Docker Hub or any registry.
2	docker push	Push an image to a Docker registry.	`docker push username/my-app`	Upload your image to Docker Hub or registry.
3	docker run	Run a container from an image.	`docker run -d -p 8080:80 --name my-nginx nginx`	Create and run a new container with port mapping and name.
4	docker stop	Stop a running container.	`docker stop my-nginx`	Stop a running container by name or ID.
5	docker start	Start an existing stopped container.	`docker start my-nginx`	Start a stopped container without creating a new one.
6	docker rm	Remove a stopped container.	`docker rm my-nginx`	Remove a container that is no longer needed.
7	docker rmi	Remove an image from local storage.	`docker rmi nginx`	Delete unused images to free up space.
8	docker ps	List running containers.	`docker ps`	Show only the currently running containers.
9	docker ps -a	List all containers (running and stopped).	`docker ps -a`	Show all containers, including stopped ones.
10	docker images	List all images on the host.	`docker images`	View all local Docker images.
11	docker exec -it	Access a running container's terminal.	`docker exec -it my-nginx bash`	Open a shell inside a running container.
12	docker build -t	Build an image from a Dockerfile.	`docker build -t my-app .`	Build a new image from Dockerfile in current directory.
13	docker logs	View the logs of a container.	`docker logs my-nginx`	Check container logs for troubleshooting.
14	docker inspect	Inspect a container or image in detail.	`docker inspect my-nginx`	Get detailed information about container/image configuration.